FramSat-1 Downlink Reception & Telemetry Decoding Guide

Technical Specifications for Amateur Radio Operators

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Abstract

This document provides the radio frequency (RF), modulation, framing, and telemetry specifications for the UHF downlink of the **FramSat-1** CubeSat, developed by Orbit NTNU. It is intended to assist amateur radio operators, SatNOGS ground station operators, and enthusiasts in receiving, demodulating, and decoding telemetry transmitted by the satellite.

1 Introduction

FramSat-1 is a 1U student satellite developed by Orbit NTNU in Trondheim, Norway. The satellite transmits periodic beacon and telemetry frames on the amateur satellite 70 cm band. All transmissions adhere strictly to open amateur radio conventions, utilizing direct synchronous HDLC-framed packets modulated with standard G3RUH 9600 bps GFSK.

Amateur radio operators are warmly invited to listen for downlinks and submit reception reports to the Orbit NTNU ground operations team.

2 Physical Layer (RF & Modulation)

The physical layer parameters for the UHF downlink are summarized in Table 1.

Table 1: FramSat-1 Downlink RF Specifications

Parameter	**Value / Specification**
Downlink Frequency	**435.141 MHz** (IARU Coordinated)
Frequency Band	70 cm Amateur Satellite Service (435–438 MHz)
Modulation Scheme	GFSK (2-FSK with Gaussian filter)
Bitrate / Baudrate	9600 bps
Frequency Deviation (Δf)	± 3.0 kHz
Modulation Index (h)	≈ 0.625
Pulse Shaping Filter	Gaussian ($BT = 0.5$, G3RUH standard)
Scrambling Scheme	Standard G3RUH ($1 + x^{12} + x^{17}$, Mask: `0x21`)
Bit Encoding	NRZI (Non-Return-to-Zero Inverted)
Preamble Length	50 bytes, nominal (`0x7E` / alternating synchronization bytes)
Postamble Length	7 bytes

3 Data Link Layer & Frame Structure

FramSat-1 utilizes synchronous HDLC framing with standard flag delimiters (`0x7E`) and a 16-bit CRC-CCITT Frame Check Sequence (FCS).

Downlink frames are transmitted directly from the onboard computer without AX.25 callsign address encapsulation. The transmission begins with a 6-byte transport/network header, followed immediately by the FramSat-1 housekeeping telemetry payload (beginning with the static ASCII signature `"FS1.0"`). The over-the-air frame layout is summarized in Table 2.

Table 2: FramSat-1 Over-The-Air Frame Structure

Byte Range	Length	Field	Description
0–5	6 B	Transport Header	Onboard network routing header (`0x85 0x00 0x85 0x00 0x00 0xCF`)
6–10	5 B	Signature	Static synchronization signature: `"FS1.0"`
11–204	194 B	Housekeeping Data	System telemetry, boot counter, battery, and sensor payload arrays
205–208	4 B	CRC / Trailer	Frame validation sequence

3.1 Frame Synchronization

Receivers should search for the 5-byte ASCII sequence `"FS1.0"` (`0x46 0x53 0x31 0x2E 0x30`). In nominal downlinks, this signature appears at byte index 6 following deframing. The reference Python decoder dynamically locks onto this signature to reliably extract telemetry regardless of transport encapsulation.

4 GNU Radio Companion (GRC) Receiver Pipeline

A complete demodulation chain can be built in GNU Radio Companion utilizing the standard `gr-satellites` and `gr-osmosdr` out-of-tree modules (both included by default in the recommended **Radioconda** distribution). Figure 1 illustrates the recommended flowgraph utilizing a standard RTL-SDR receiver.

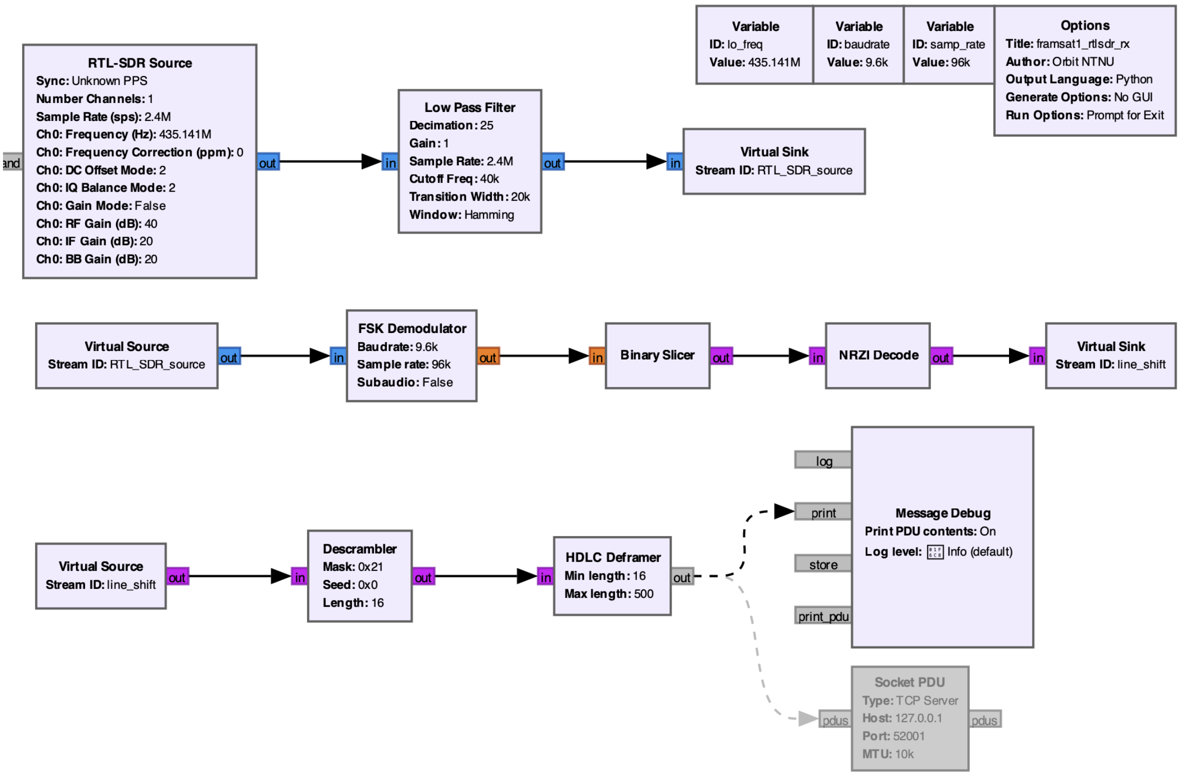

Figure 1: Recommended GNU Radio receiver pipeline for FramSat-1 using an RTL-SDR dongle (sampling at 2.4 MSps and decimated to 96 kSps).

The recommended demodulator architecture consists of:

1. **RTL-SDR Source:** Center frequency set to 435.141 MHz, sample rate 2.4 MSps, and RF Gain set to 40 dB.
2. **Low Pass Filter (Decimator):** Decimates by a factor of 25 (reducing the rate from 2.4 MSps to 96 kSps), with a cutoff frequency of 40 kHz and transition width of 20 kHz to suppress out-of-band noise.
3. **FSK Demodulator:** Operates at the decimated sample rate of 96 kSps with baud rate set to 9600 bps.
4. **Binary Slicer:** Converts demodulated audio-frequency excursions into binary symbols (0 and 1).
5. **NRZI Decode:** Decodes bit transitions according to standard G3RUH / NRZI conventions.
6. **Descrambler:** Mask: `0x21`, Seed: `0x0`, Length: `16` (Standard G3RUH polynomial).
7. **HDLC Deframer:** Minimum frame length: 16 bytes, Maximum: 500 bytes. Validates the 16-bit CRC-CCITT (FCS) and discards corrupt frames.
8. **Message Debug / Socket PDU:** Outputs valid received frames to the console via `Message Debug`, or optionally streams them to external software via `Socket PDU` (TCP Server on port 52001).

5 Reference Frame Decoder

To assist the amateur radio community, an open-source Python reference decoder is maintained in the project repository:

https://github.com/orbitntnu/framsat1-telemetry-decoder

The script (`framsat1_decoder.py`) provides two operational modes:

- **Live Network Mode (`--listen`):** Connects directly to GNU Radio via TCP (port 52001) to ingest, deframe, and display satellite telemetry in real time during an active pass.
- **Offline / Inspection Mode:** Accepts a raw hexadecimal frame string as a command-line argument to parse historical or crowdsourced frames recorded by other stations.

The decoder dynamically identifies the `"FS1.0"` synchronization signature (at byte offset 6 or 0), verifies frame integrity, and unpacks the full housekeeping telemetry structure into human-readable engineering units.

6 Housekeeping Telemetry Format

FramSat-1 transmits periodic health and status telemetry within the received frame, beginning with the 5-byte ASCII signature `"FS1.0"` (located at byte offset 6 following the 6-byte transport header). All multi-byte numeric fields are serialized in standard **Big-Endian** format (network byte order, MSB first). The layout of the telemetry payload is specified in Table 3 (offsets are relative to the `"FS1.0"` signature):

Table 3: FramSat-1 Housekeeping Telemetry Format

Offset	Size	Field	Type	Description / Scaling
0–4	5 B	`signature`	ASCII String	Static sync signature: `"FS1.0"`
5	1 B	`beacon_id`	`uint8_t`	Packet sequence counter
6	1 B	`operating_mode`	`uint8_t`	Mode: 1 = Default, 2 = LEOP
7	1 B	`satellite_rssi`	`uint8_t`	Ground uplink RSSI recorded by satellite
8–9	2 B	`telecommand_count`	`uint16_t`	Total telecommands accepted by satellite
10	1 B	`eps_flags`	`uint8_t`	EPS status and subsystem power flags
11–12	2 B	`reboot_count`	`uint16_t`	Total OBC / EPS reboot counter
13–14	2 B	`battery_voltage`	`uint16_t`	Battery voltage ($V = \text{val}/1000.0\,\text{V}$)
15–18	4 B	`uptime_sec`	`uint32_t`	Satellite uptime in seconds
19–198	180 B	`sensor_data`	Array[9]	9x Attitude sensor samples (Sun & Earth sensors)

6.1 Identification Signature

Ground stations can reliably identify FramSat-1 downlinks by verifying that the received payload contains the ASCII signature `"FS1.0"` (at byte offset 6 of the deframed data).

7 Submitting Reception Reports

Orbit NTNU greatly appreciates reports and raw telemetry recordings from the radio community. If you receive packets from FramSat-1, please submit:

- Your callsign and ground station location (grid square or coordinates).
- UTC timestamp of reception.
- Raw hexadecimal packet string or IQ baseband recording (`.wav` / `.sigmf`).
- Estimated SNR or waterfall screenshot.

Reports can be submitted via email to satcom@orbitntnu.com or uploaded to the SatNOGS network database under satellite **FramSat-1**.